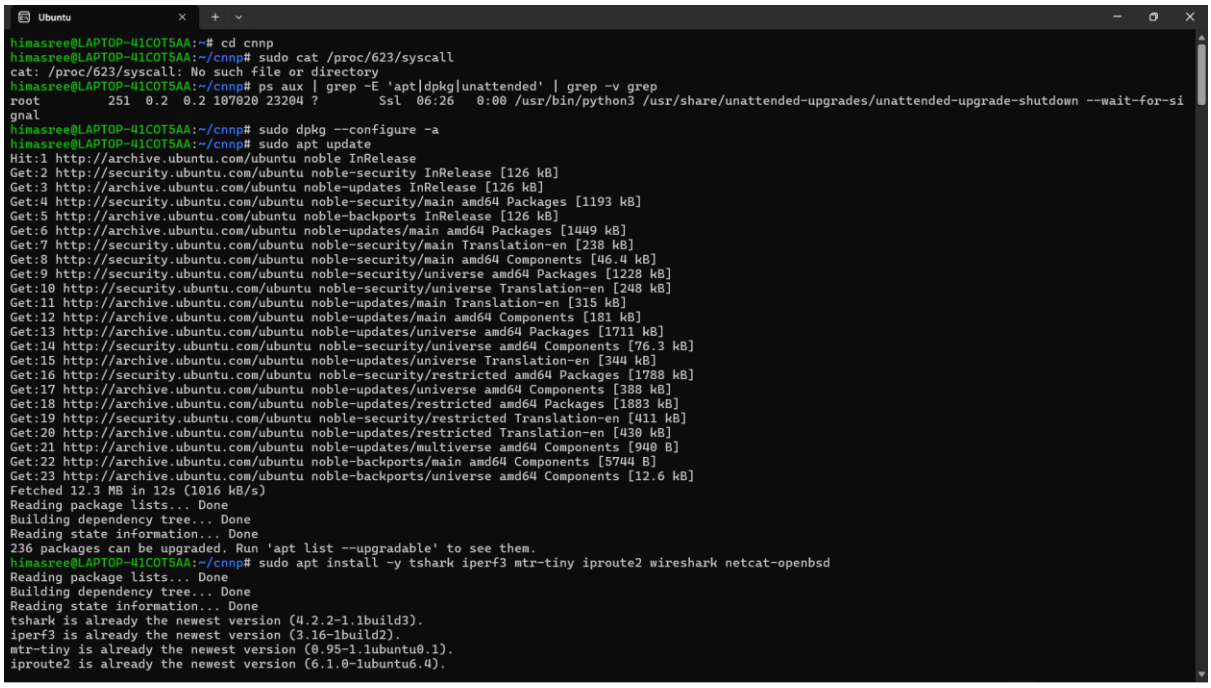
Surapuraju Himasree

2520090181

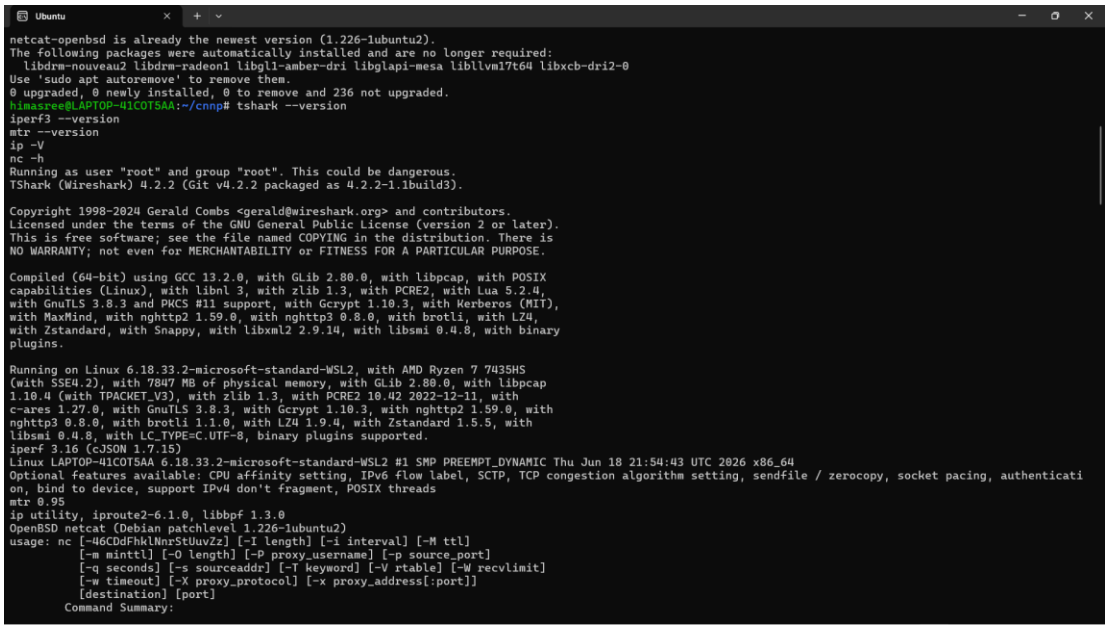
Sec – 11

Skill – 2

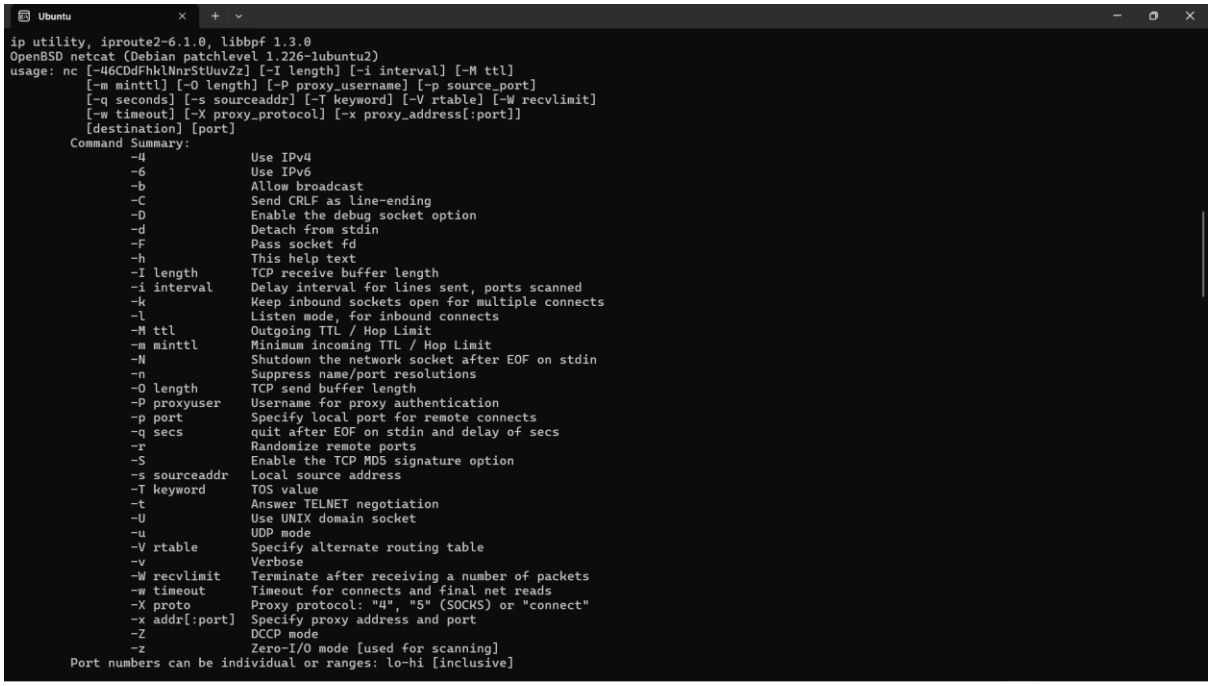
Step - 1 : Open Ubuntu and install the required networking and packet-analysis tools, including TShark, iperf3, mtr, iproute2, Wireshark and Netcat.



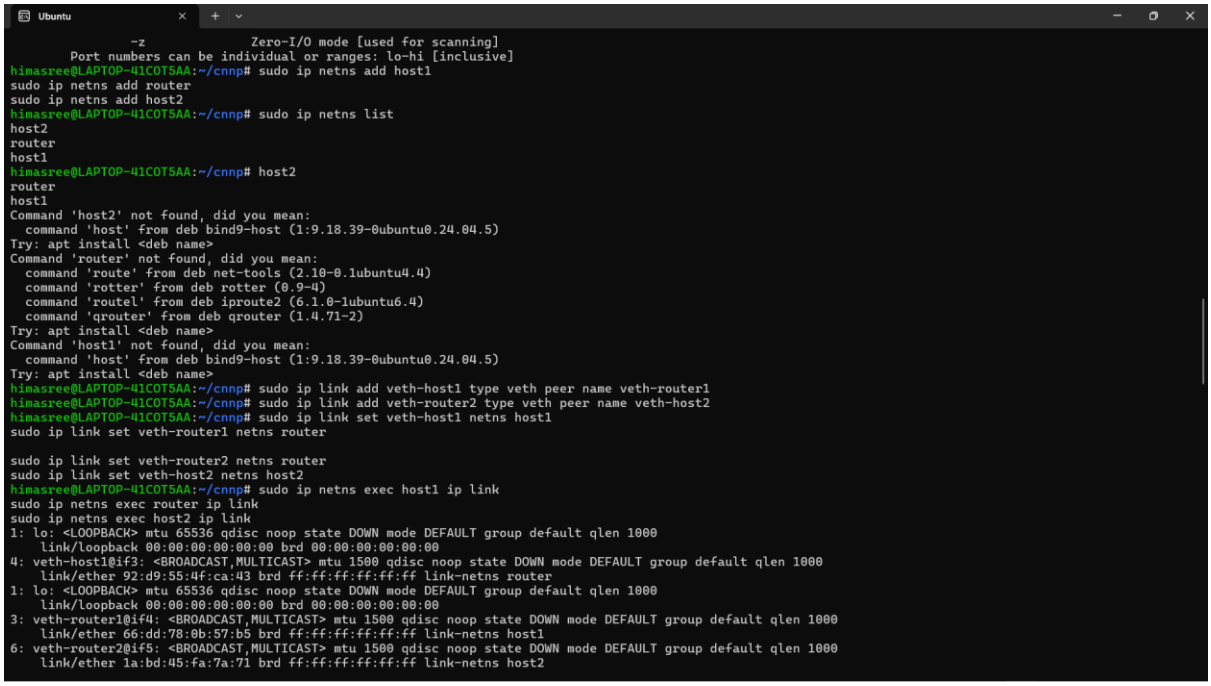
Step - 2 : Create the three network namespaces named host1, router and host2 to represent the sender, intermediate router and receiver.



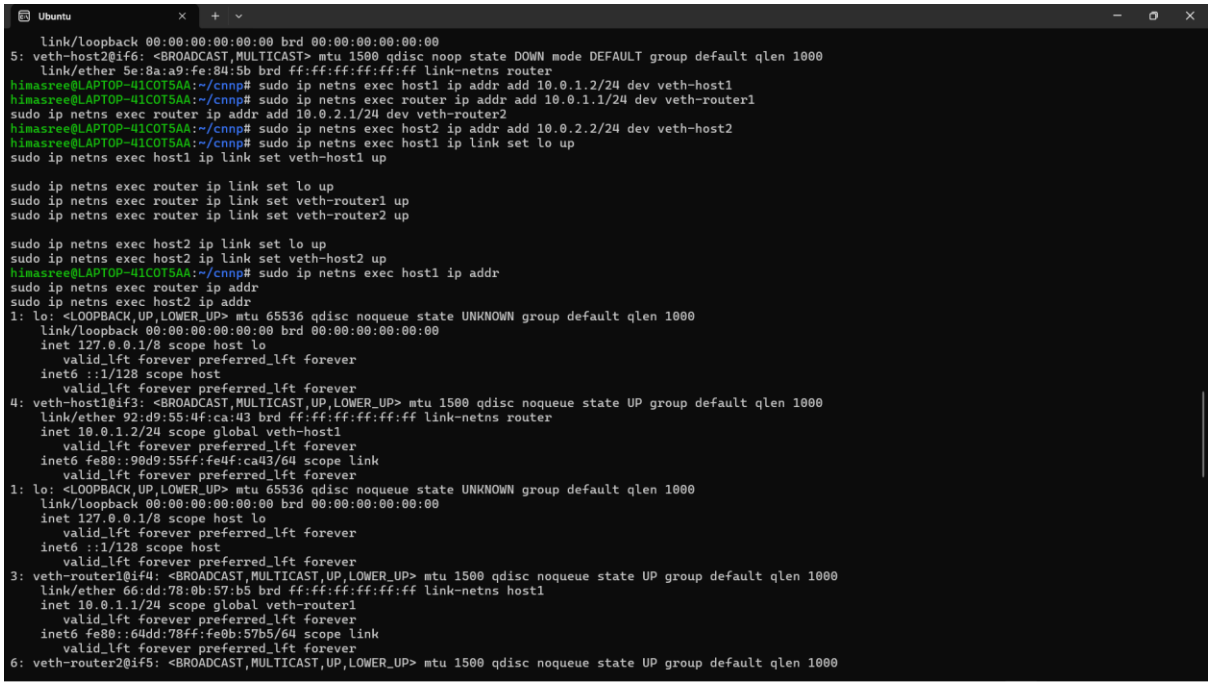
Step - 3 : Create two virtual Ethernet cable pairs (veth pairs) to connect host1 to the router and the router to host2.



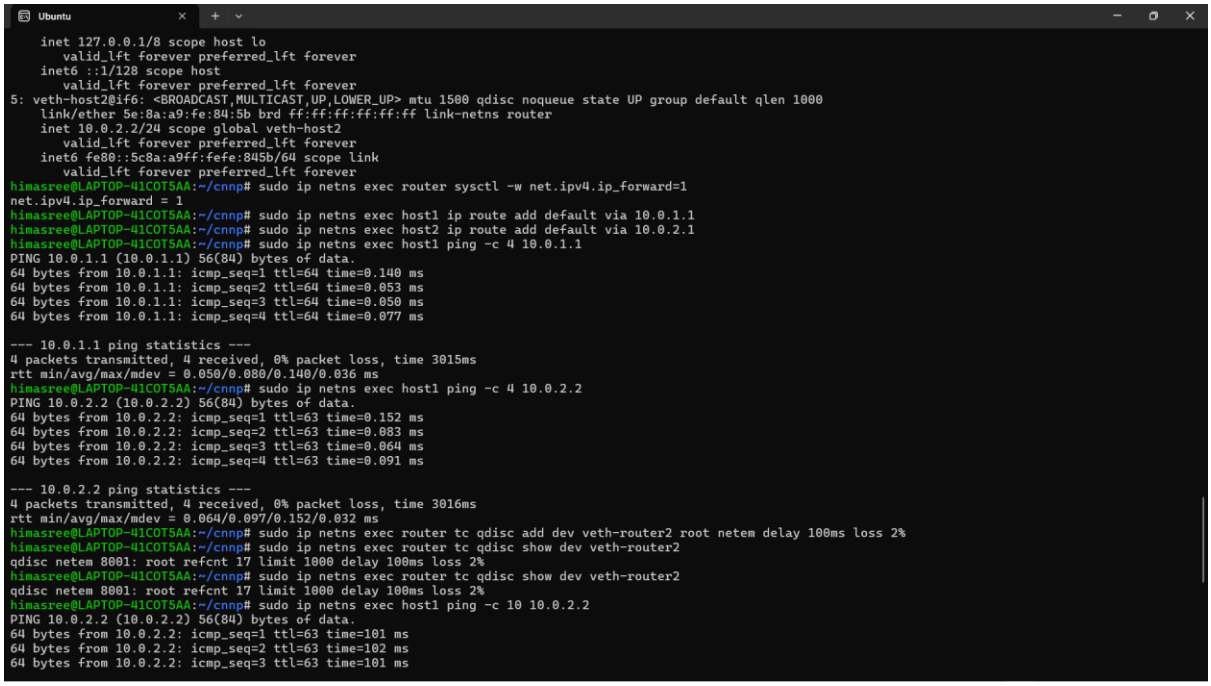
Step - 4 : Move each virtual Ethernet interface into its corresponding network namespace so that the three nodes are connected through the router.



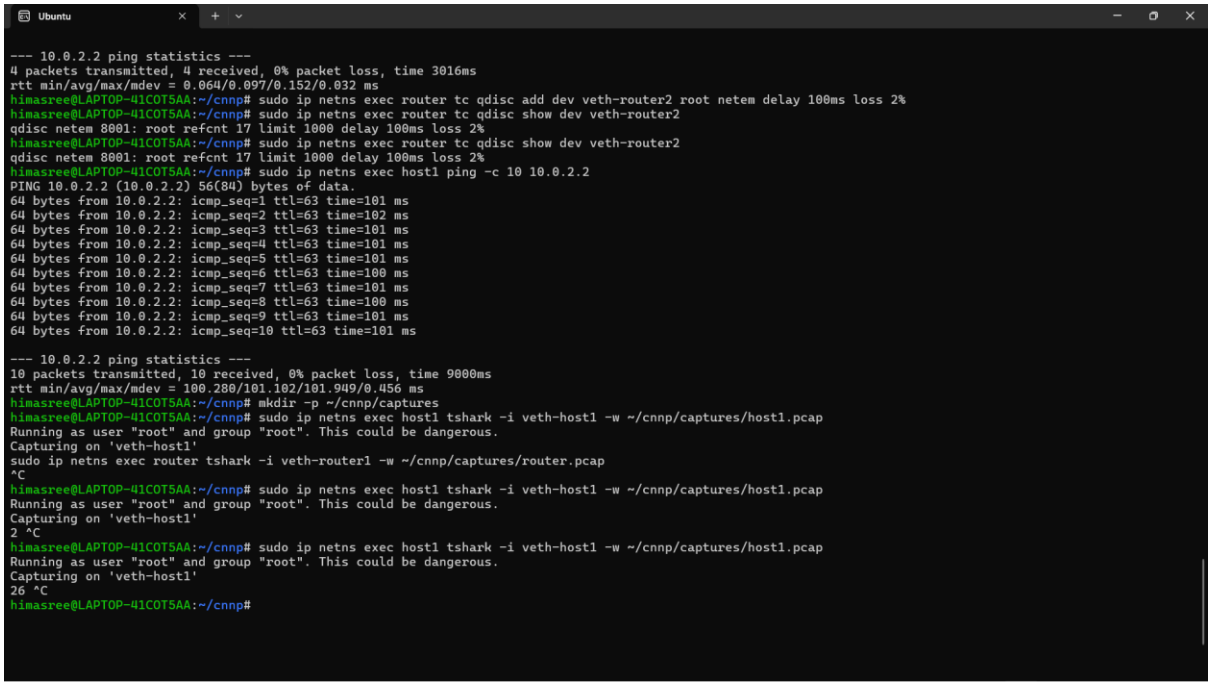
Step - 5 : Configure IP addresses on host1, both router interfaces and host2, then bring up the interfaces and loopback devices.



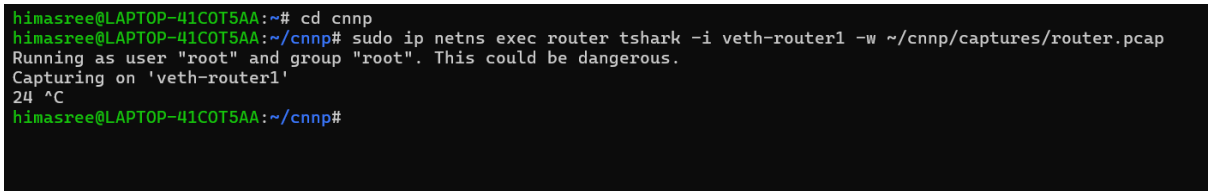
Step - 6 : Enable IPv4 forwarding on the router and add the routes on host1 and host2 so packets can travel between the two networks.



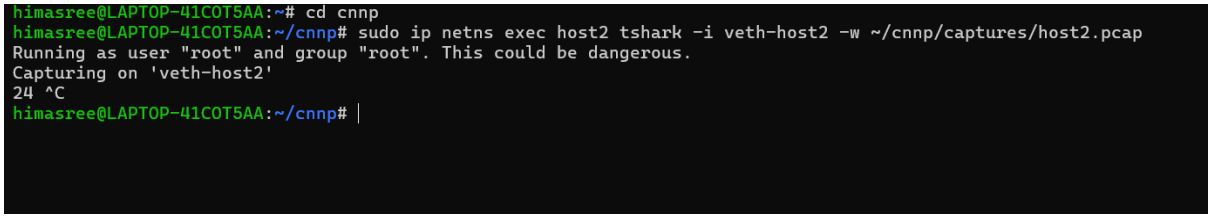
Step - 7 : Test end-to-end connectivity by sending ping packets from host1 to host2 and verify that the packets are successfully received.



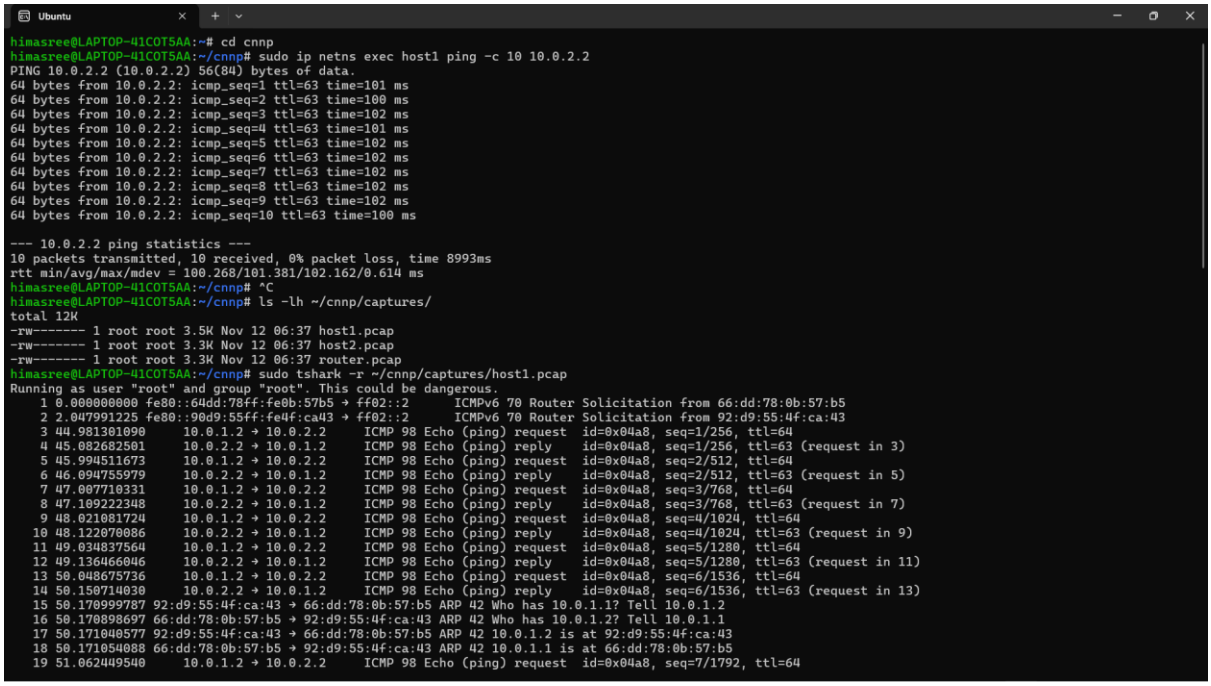
Step - 8 : Add network emulation on the router interface with 20 ms delay and 1% packet loss to simulate real network conditions.



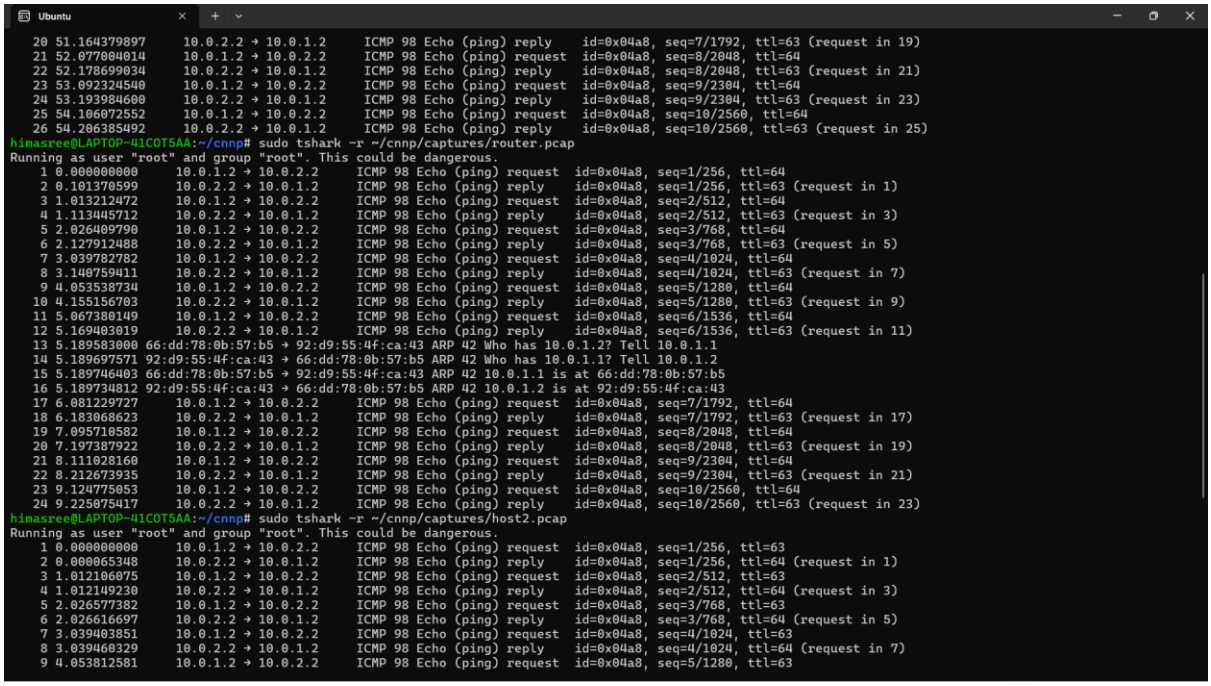
Step - 9 : Verify the configured netem delay and packet-loss settings on the router interface before starting packet capture.



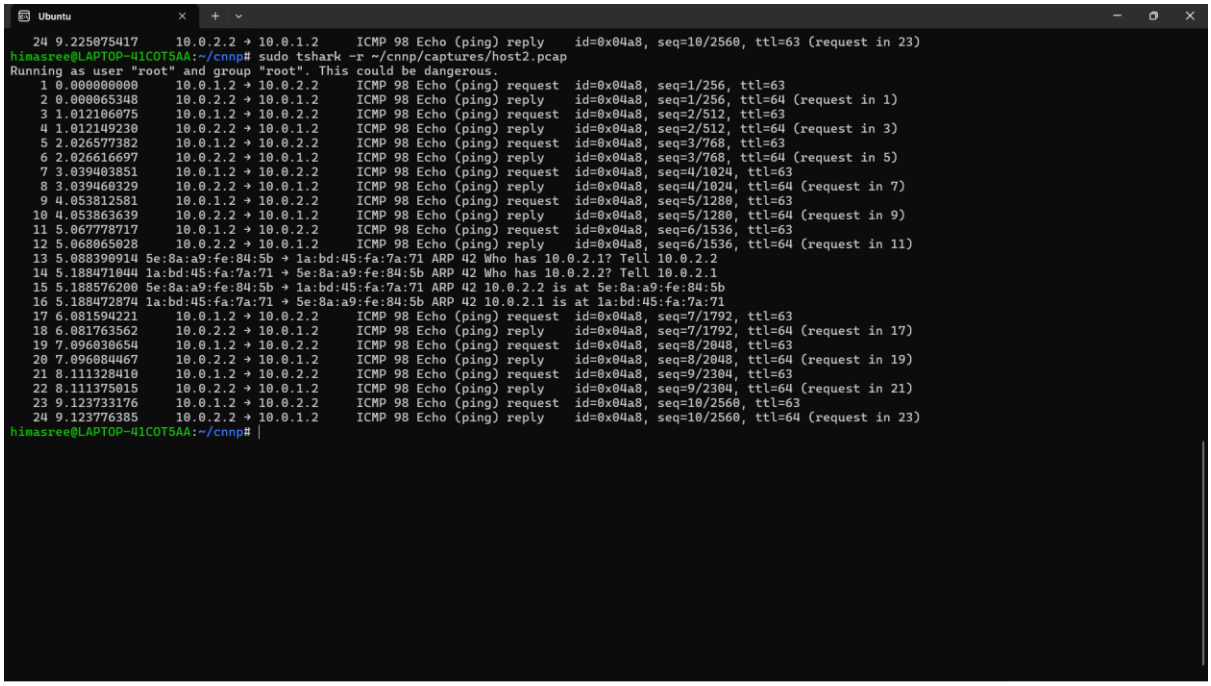
Step - 10 : Open the required terminals and start the first packet capture on host1 using TShark, saving the packets to a capture file.



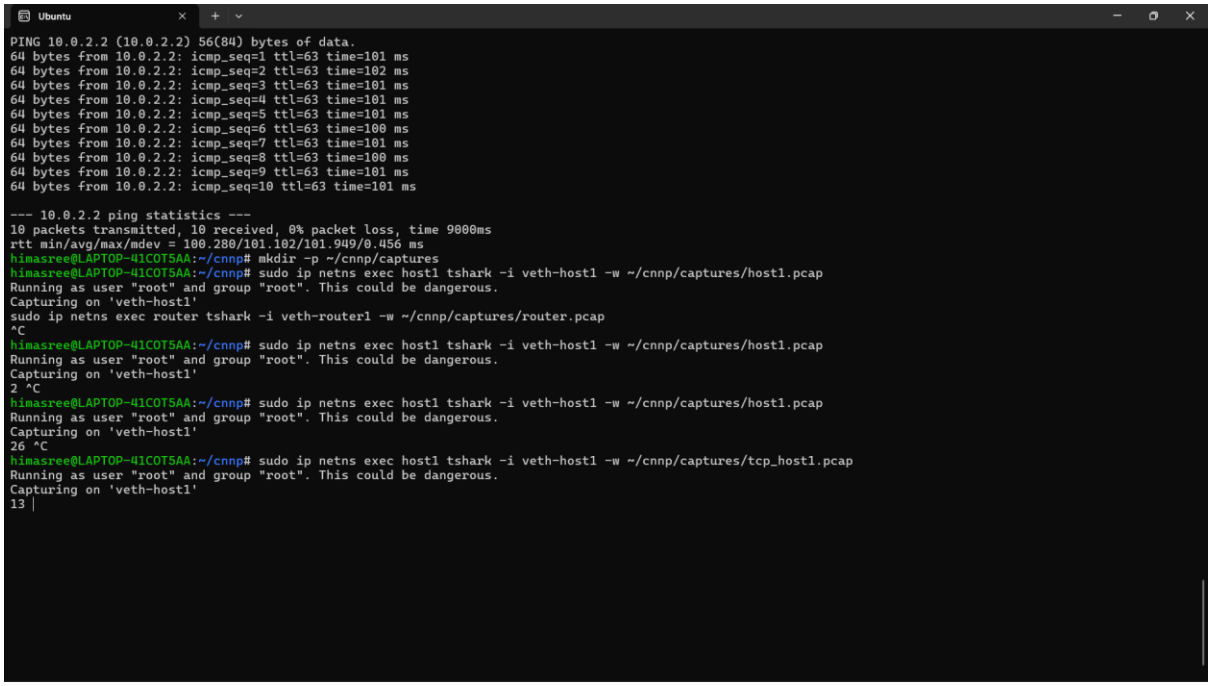
Step - 11 : Start the second simultaneous packet capture on the router interface to observe packets passing through the intermediate node.



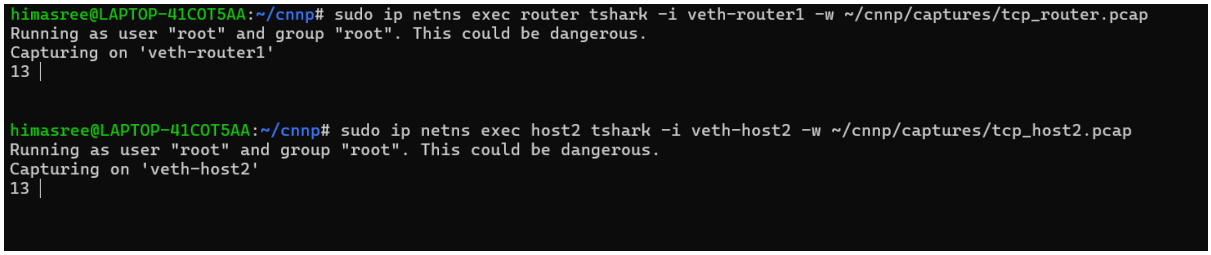
Step - 12 : Start the third simultaneous packet capture on host2 to observe packets arriving at the destination.



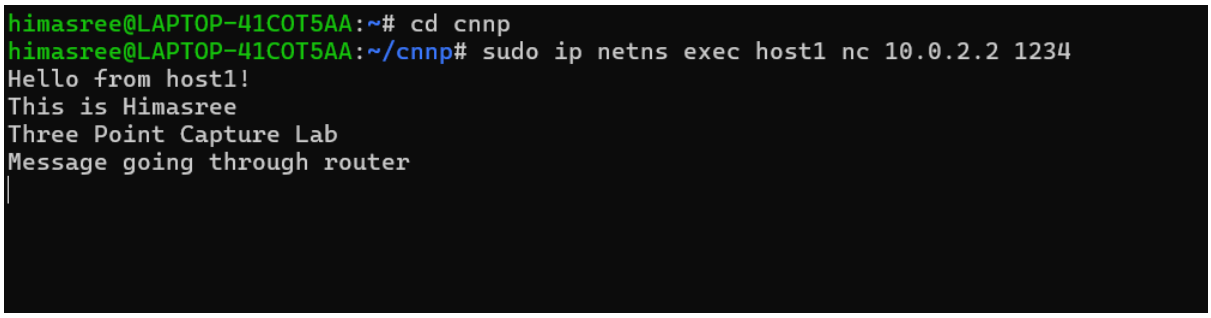
Step - 13 : Keep all three TShark capture sessions running simultaneously so the same network traffic is recorded at all three points.



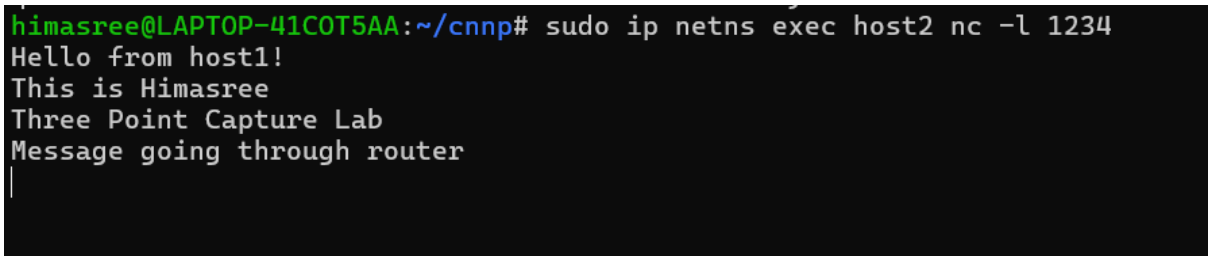
Step - 14 : Start the Netcat receiver on host2 and connect to it from host1 using TCP port 1234 to establish the message-transfer connection.



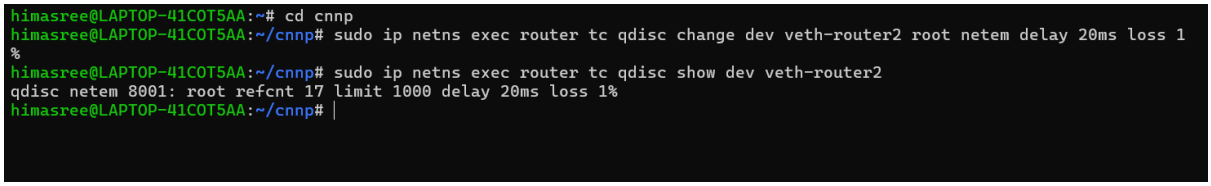
Step - 15 : Send the sample text message from host1 through the router to host2 and confirm that the receiver displays the transmitted message.



Step - 16 : Check the receiving terminal and verify that the complete message sent by host1 has arrived successfully at host2.



Step - 17 : Stop the packet-capture sessions after the message transfer so that the three capture files are saved for analysis.



Step - 18 : List the saved capture files and check that the host1, router and host2 PCAP files exist and contain captured data.

```
himasree@LAPTOP-41COT5AA:~/cnnp# ls -lh ~/cnnp/captures/
total 24K
-rw------- 1 root root 3.5K Nov 12 06:37 host1.pcap
-rw------- 1 root root 3.3K Nov 12 06:37 host2.pcap
-rw------- 1 root root 3.3K Nov 12 06:37 router.pcap
-rw------- 1 root root 2.2K Nov 12 07:00 tcp_host1.pcap
-rw------- 1 root root 1.9K Nov 12 07:00 tcp_host2.pcap
-rw------- 1 root root 2.2K Nov 12 07:00 tcp_router.pcap
himasree@LAPTOP-41COT5AA:~/cnnp#
```

Step - 19 : Read the IP Time to Live (TTL) values from each capture using TShark to compare packet values at the three observation points.

```
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_host1.pcap -Y tcp
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000     10.0.1.2 → 10.0.2.2     TCP 74 58692 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1900064758 TSecr=0 WS=1024
    2 0.020215193     10.0.2.2 → 10.0.1.2     TCP 74 1234 → 58692 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2830781309 TSecr=190006475
8 WS=1024
    3 0.020275137     10.0.1.2 → 10.0.2.2     TCP 66 58692 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=1900064779 TSecr=2830781309
    8 9.908959429     10.0.1.2 → 10.0.2.2     TCP 84 58692 → 1234 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=18 TSval=1900074667 TSecr=2830781309
    9 9.929299021     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [ACK] Seq=1 Ack=19 Win=65536 Len=0 TSval=2830791218 TSecr=1900074667
   10 9.929324039     10.0.1.2 → 10.0.2.2     TCP 107 58692 → 1234 [PSH, ACK] Seq=19 Ack=1 Win=64512 Len=41 TSval=1900074688 TSecr=2830791218
   11 9.949598690     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [ACK] Seq=1 Ack=60 Win=65536 Len=0 TSval=2830791238 TSecr=1900074688
   12 12.856268886     10.0.1.2 → 10.0.2.2     TCP 95 58692 → 1234 [PSH, ACK] Seq=60 Ack=1 Win=64512 Len=29 TSval=1900077615 TSecr=2830791238
   13 12.876447247     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=2830794165 TSecr=1900077615
   14 251.578195044     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [FIN, ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=2831032866 TSecr=1900077615
   15 251.582100092     10.0.1.2 → 10.0.2.2     TCP 66 58692 → 1234 [ACK] Seq=89 Ack=2 Win=64512 Len=0 TSval=1900316341 TSecr=2831032866
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_router.pcap -Y tcp
Running as user "root" and group "root". This could be dangerous.
    1 0.000000000     10.0.1.2 → 10.0.2.2     TCP 74 58692 → 1234 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=1900064758 TSecr=0 WS=1024
    2 0.020197259     10.0.2.2 → 10.0.1.2     TCP 74 1234 → 58692 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2830781309 TSecr=190006475
8 WS=1024
    3 0.020267763     10.0.1.2 → 10.0.2.2     TCP 66 58692 → 1234 [ACK] Seq=1 Ack=1 Win=64512 Len=0 TSval=1900064779 TSecr=2830781309
    8 9.908961153     10.0.1.2 → 10.0.2.2     TCP 84 58692 → 1234 [PSH, ACK] Seq=1 Ack=1 Win=64512 Len=18 TSval=1900074667 TSecr=2830781309
    9 9.929277440     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [ACK] Seq=1 Ack=19 Win=65536 Len=0 TSval=2830791218 TSecr=1900074667
   10 9.929321404     10.0.1.2 → 10.0.2.2     TCP 107 58692 → 1234 [PSH, ACK] Seq=19 Ack=1 Win=64512 Len=41 TSval=1900074688 TSecr=2830791218
   11 9.949578282     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [ACK] Seq=1 Ack=60 Win=65536 Len=0 TSval=2830791238 TSecr=1900074688
   12 12.856272079     10.0.1.2 → 10.0.2.2     TCP 95 58692 → 1234 [PSH, ACK] Seq=60 Ack=1 Win=64512 Len=29 TSval=1900077615 TSecr=2830791238
   13 12.876429194     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=2830794165 TSecr=1900077615
   14 251.578174025     10.0.2.2 → 10.0.1.2     TCP 66 1234 → 58692 [FIN, ACK] Seq=1 Ack=89 Win=65536 Len=0 TSval=2831032866 TSecr=1900077615
   15 251.582099721     10.0.1.2 → 10.0.2.2     TCP 66 58692 → 1234 [ACK] Seq=89 Ack=2 Win=64512 Len=0 TSval=1900316341 TSecr=2831032866
himasree@LAPTOP-41COT5AA:~/cnnp#
```

Step - 20 : Extract the TCP payload from the router capture and verify that the actual message sent from host1 is present in the captured packets.

```
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_router.pcap -Y "tcp.port == 1234" -T fields -e tcp.payload 2>/dev/null | xxd | head -20
00000000: 0a0a 0a34 3836 3536 6336 6632 3036  ...48656c6c6f206
00000010: 3637 3236 6636 6432 3036 3836 6637 3337  6726f6d20686f737
00000020: 3433 3132 3130 610a 0a35 3436 3836 3937  431210a..5468697
00000030: 3332 3036 3937 3332 3034 3836 3936 6436  3206973204869 6d6
00000040: 3137 3337 3236 3536 3530 6135 3436 3837  1737265650a54687
00000050: 3236 3536 3532 3035 3036 6636 3936 6537  2656520506f696e7
00000060: 3432 3034 3336 3137 3037 3437 3537 3236  4204361707475726
00000070: 3532 3034 6336 3136 3230 610a 0a34 6436  5204c61620a..4d6
00000080: 3537 3337 3336 3136 3736 3532 3036 3736  5737361676520676
00000090: 6636 3936 6536 3732 3037 3436 3837 3236  f696e67207468726
000000a0: 6637 3536 3736 3832 3037 3236 6637 3537  f75676820726f757
000000b0: 3436 3537 3230 610a 0a0a 0a              465720a....
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_router.pcap -Y "tcp.port == 1234 && tcp.len > 0" -T fields -e tcp.payload 2>/dev/null |
tr -d '\n' | xxd -r -p
Hello from host1!
This is Himasree
Three Point Capture Lab
Message going through router
himasree@LAPTOP-41COT5AA:~/cnnp#
```

Step - 21 : Compare the relative timestamps of the TCP SYN packet in the three captures to observe the delay introduced along the network path.

```
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_host1.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      6 64
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_router.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      5 63
      6 64
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_host2.pcap -T fields -e ip.ttl -Y "ip" 2>/dev/null | sort | uniq -c
      6 63
      5 64
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec host1 ping -c 5 10.0.2.2
PING 10.0.2.2 (10.0.2.2) 56(84) bytes of data.
64 bytes from 10.0.2.2: icmp_seq=1 ttl=63 time=21.1 ms
64 bytes from 10.0.2.2: icmp_seq=2 ttl=63 time=20.4 ms
64 bytes from 10.0.2.2: icmp_seq=3 ttl=63 time=21.1 ms
64 bytes from 10.0.2.2: icmp_seq=4 ttl=63 time=21.1 ms
64 bytes from 10.0.2.2: icmp_seq=5 ttl=63 time=20.2 ms

--- 10.0.2.2 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 3998ms
rtt min/avg/max/mdev = 20.182/20.789/21.143/0.410 ms
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_host1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
himasree@LAPTOP-41COT5AA:~/cnnp#
```

Step - 22 : Check the TCP retransmission counts in all three capture files and run an MTR report from host1 to host2 to inspect hop-wise delay and packet loss.

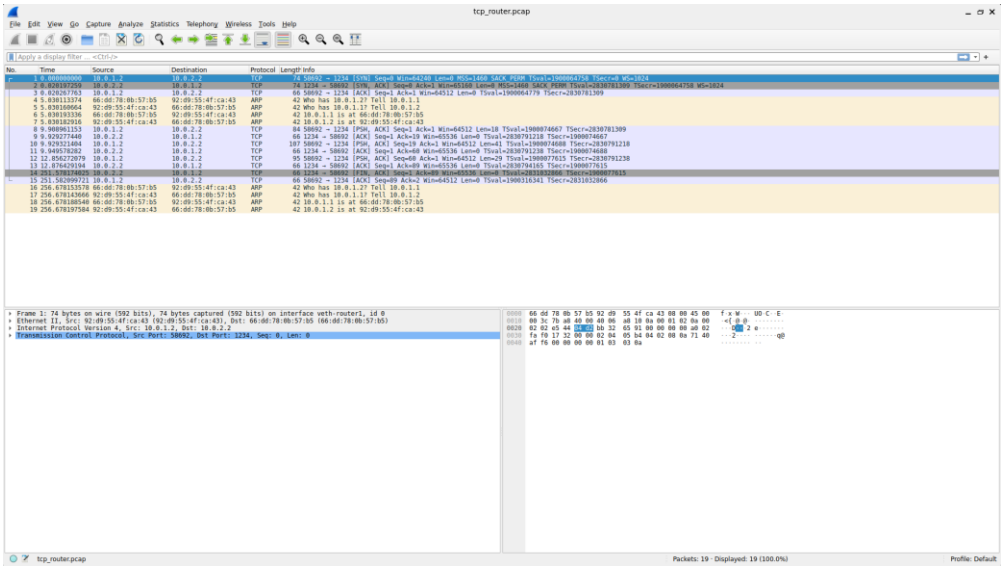
```
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_host1.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_router.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
himasree@LAPTOP-41COT5AA:~/cnnp# sudo tshark -r ~/cnnp/captures/tcp_host2.pcap -Y "tcp.analysis.retransmission" 2>/dev/null | wc -l
0
himasree@LAPTOP-41COT5AA:~/cnnp# sudo ip netns exec host1 mtr --report 10.0.2.2
Start: 2026-11-12T07:06:05+0000
HOST: LAPTOP-41COT5AA             Loss%   Snt   Last   Avg  Best  Wrst StDev
  1.|-- 10.0.1.1                   0.0%    10    0.1   0.1   0.1   0.5   0.1
  2.|-- 10.0.2.2                   0.0%    10   20.2  20.4  20.1  21.0   0.3
himasree@LAPTOP-41COT5AA:~/cnnp# wireshark ~/cnnp/captures/tcp_router.pcap &
[1] 1456
himasree@LAPTOP-41COT5AA:~/cnnp# libEGL warning: failed to get driver name for fd -1

libEGL warning: MESA-LOADER: failed to retrieve device information

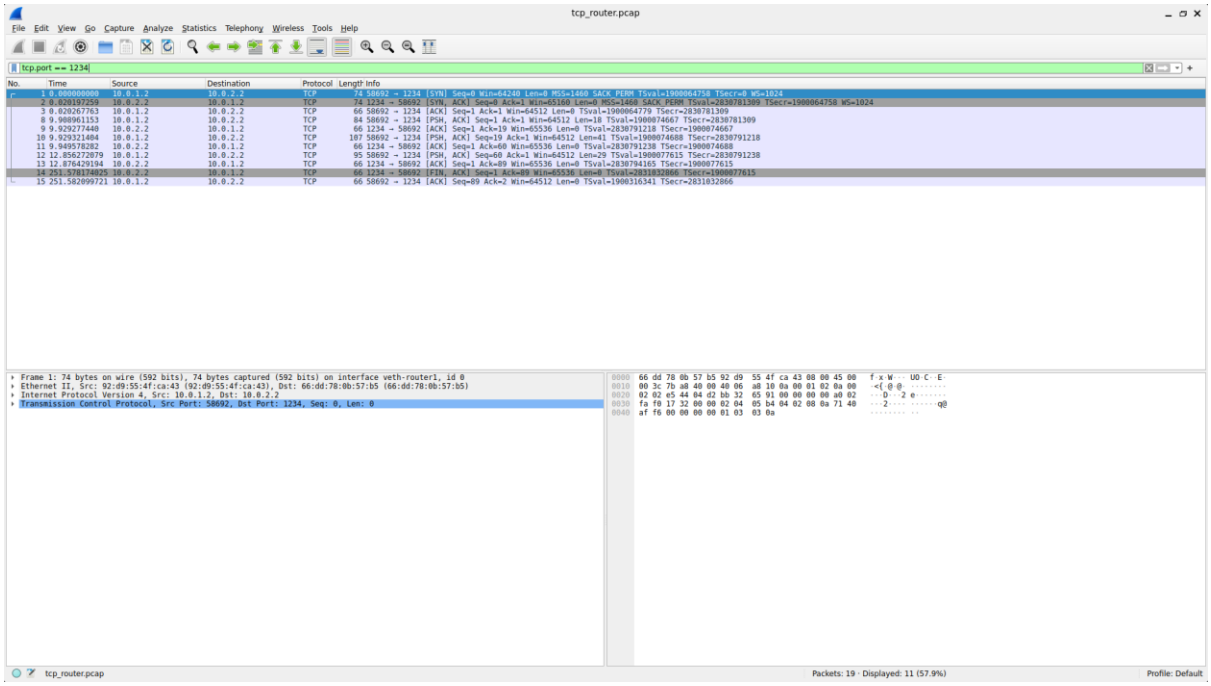
libEGL warning: failed to get driver name for fd -1

MESA: error: ZINK: failed to choose pdev
libEGL warning: egl: failed to create dri2 screen
```

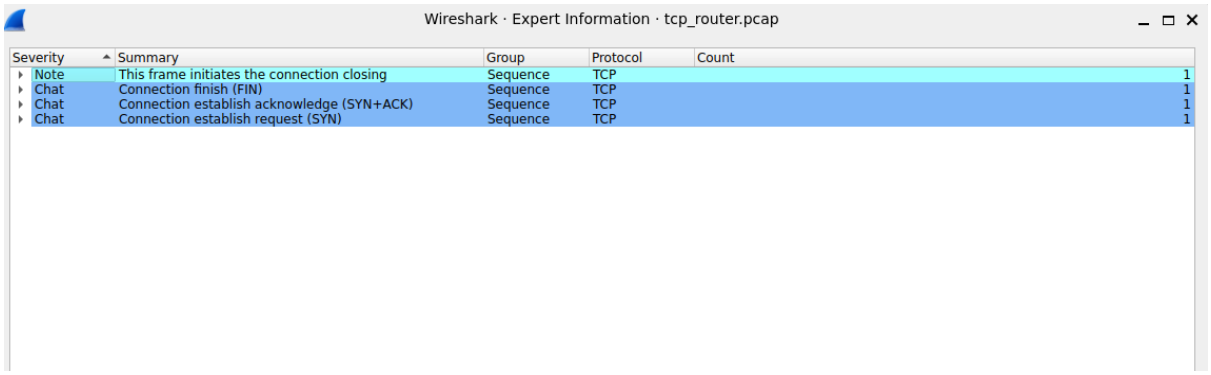
Step - 23 : Open the router capture in Wireshark and apply the display filter tcp.port == 1234 to view the TCP communication for the message transfer.



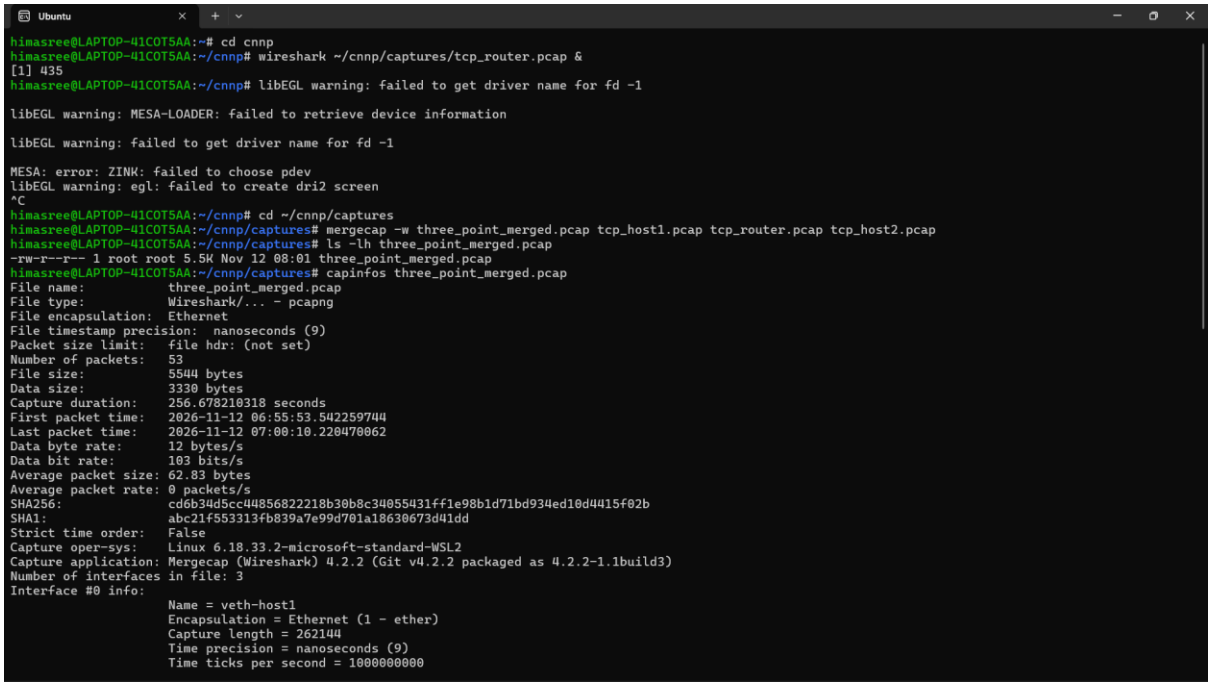
Step - 24 : Inspect the selected packet details in Wireshark and expand Ethernet II, IPv4, TCP and Data to view the headers, TTL value and message payload.



Step - 25 : Merge the three packet captures in Wireshark and add a custom TTL column using the ip.ttl field to compare TTL values across capture points.



Step - 26 : Open Analyze → Expert Information in Wireshark to review packet-analysis events and identify any reported network or TCP issues.



Step - 27 : Clean up the experiment by deleting the host1, router and host2 network namespaces, then verify that the namespace list is empty.

